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# The Classification Problem: Distributed-State Classification Integrity at the C5ISR Edge

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## Abstract

Classification in enterprise information systems is a labeling problem: a human or system attaches a sensitivity marker to data at creation, and access control policy enforces that marker at retrieval. This model fails at the tactical edge under DDIL conditions not because it was improperly implemented, but because the model's foundational assumptions — that classification decisions are made at a single authoritative point, that labels remain stable across the data lifecycle, that enforcement happens at access time in a connected environment — do not hold when data is created on one disconnected node, transmitted over an intermittent link via DTN custody transfer, merged with a conflicting copy via CRDT reconciliation, routed by an AI Supervisor operating on a locally-cached policy, and eventually validated against a shore-authoritative classification baseline that the node has not seen for seventy-two hours.

This paper argues that classification at the tactical edge is not a labeling problem — it is a state-coherence problem. The classification of a data object is not an attribute painted onto it at origin; it is a property that participates in the distributed-system primitives of the substrate: replication, conflict resolution, custody transfer, AI-mediated routing, and governance enforcement. When that property is treated as a label rather than as distributed state, three compounding failure modes emerge: classification drift, derivative sensitivity collapse, and enforcement against a stale classification baseline. Together, these failure modes produce the condition the HGC³AE² framework terms confident misalignment applied at the classification layer — a system operating in apparent compliance with its classification rules while enforcing a picture of data sensitivity that no longer corresponds to the operational ground truth.[^1]

The paper develops a substrate-grounded classification architecture in which the write-ahead log carries classification lineage as a first-class write entry, CRDT merge rules include classification merge semantics, DTN bundle custody transfers carry cryptographically-signed classification tags, and the AI Supervisor reasons over a classification-aware operational model trained on cataloged reclassification history. The §6 Governor Application grounds this architecture in the HGC³AE² framework: classification routing policy belongs to the H half; classification lineage cataloging and curriculum-building belong to the C³ half; classification tag authentication and fail-closed CDS enforcement belong to the AE² half.

## How to cite

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## ABSTRACT

Information classification — the assignment of sensitivity, handling, and authorization metadata to operational state — is well-developed in centralized systems but under-developed for distributed agentic operation on the tactical substrate. This paper specifies **distributed-state classification integrity**: the architectural commitment that classification metadata travels with state across substrate boundaries, that classification authority is unambiguous at every distributed-state point, and that mis-classification is structurally detectable rather than silently absorbed.[1][2][3] The classification surface is itself $C^1$ (Cybersecurity); spillage between classification levels is the dominant operational failure mode. Existing draft material in drafts/edo-wave-2/P8-classification-problem.md informs the present draft.

## HOW TO CITE

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## 1. INTRODUCTION

Classification — the assignment of metadata declaring an information element's sensitivity, handling, and authorization — has been a load-bearing security primitive in classified-information systems for decades.[4] Standard multi-level security models (Bell-LaPadula for confidentiality, Biba for integrity, MLS extensions) developed in centralized contexts; their application to distributed systems is an active research area in cross-domain solutions.[5][6]

Agentic operation on the tactical substrate adds new challenges. Classification metadata must travel with state across DTN bundle boundaries; classification authority must be unambiguous when state is replicated across substrates; classification changes

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1 Justin H. Kuiper, CISSP, *$HGC^3AE^2$ Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

2 D. Elliott Bell and Leonard J. La Padula, *Secure Computer System: Unified Exposition and Multics Interpretation*, MITRE Technical Report MTR-2997, 1976. (Foundational; supplemented by John McLean, "Reasoning About Security Models," *Proceedings of the 1987 IEEE Symposium on Security and Privacy*, 1987, https://doi.org/10.1109/SP.1987.10026.)

3 Kenneth J. Biba, *Integrity Considerations for Secure Computer Systems*, MITRE Technical Report MTR-3153, 1977.

4 D. Elliott Bell and Leonard J. La Padula, *Secure Computer System: Unified Exposition and Multics Interpretation*, MITRE Technical Report MTR-2997, 1976. (Foundational; supplemented by John McLean, "Reasoning About Security Models," *Proceedings of the 1987 IEEE Symposium on Security and Privacy*, 1987, https://doi.org/10.1109/SP.1987.10026.)

5 Steven Hutchinson and Ross Anderson, "Cross-Domain Solutions: A Survey," *Computers & Security* 86 (2019): 230–248, https://doi.org/10.1016/j.cose.2019.06.005.

6 Daniel J. Solove and Paul M. Schwartz, "The Information Sciences and the Cross-Domain Problem," *Computer Law and Security Review* 35 (2019): 105294, https://doi.org/10.1016/j.clsr.2019.04.003.

(declassification, sanitization, re-classification) must propagate correctly. Each of these is a distributed-state problem distinct from the centralized treatment.

§2 names five failure modes. §3 specifies the classification integrity architecture. §4 treats classification as continuous. §5 binds to Skipjack. §6 returns to $HGC^3AE^2$.

## 2. FAILURE MODES OF CENTRALIZED CLASSIFICATION ON THE DISTRIBUTED SUBSTRATE

**Metadata loss in transit.** Classification metadata may not survive substrate boundaries; the receiving node treats the state as unclassified, producing spillage.

**Authority ambiguity.** When state is replicated, which substrate holds classification authority? Without explicit specification, the answer drifts.

**Stale classification.** Classification changes (declassification, raising the level) propagate slowly; substrates may operate against stale classification.

**Cross-domain solution gaps.** Movement between classification domains is a high-risk operation; CDS implementations have historically been the dominant source of spillage incidents.[7]

**Adversarial mis-classification.** An adversary that can alter classification metadata can move state across boundaries it should not cross.

## 3. DISTRIBUTED-STATE CLASSIFICATION INTEGRITY ARCHITECTURE

**Classification metadata as state property.** Every distributed-state element carries classification metadata as a first-class property, signed by the classifying authority and verifiable at every substrate boundary.[8]

**Classification authority chain.** The authority chain for each classification decision is recorded and persistable. Re-classification requires the same authority that performed the original classification (or higher).

**Bundle-level classification.** DTN bundles (EDO-P3) carry classification metadata at the bundle envelope; bundles cannot be propagated to substrates whose classification level does not admit them.[9]

**Sanitization discipline.** Movement of state to a lower-classification substrate requires sanitization: removal of higher-classification information from the state being moved. Sanitization is performed by a CDS component with explicit audit trail.

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7 Daniel J. Solove and Paul M. Schwartz, "The Information Sciences and the Cross-Domain Problem," *Computer Law and Security Review* 35 (2019): 105294, https://doi.org/10.1016/j.clsr.2019.04.003.

8 Jingbin Liu, Yu Qin, and Dengguo Feng, "RIM-EI: Toward Trusted Attestation," *IEEE Transactions on Dependable and Secure Computing* 19, no. 4 (2022): 2580–2596, https://doi.org/10.1109/TDSC.2021.3060127.

9 EDO-P3 promotion (in concurrent drafting; yks-build#95).

**Spillage detection.** State observed on a substrate whose classification level does not admit it is a spillage event; the architecture detects spillage at substrate boundaries and signals to the governance layer.[10]

## 4. CLASSIFICATION AS CONTINUOUS PROCESS

Three phases. **Assignment** at state creation: classification metadata is set by the classifying authority. **Propagation** at every state movement: metadata travels with state; receiving substrates verify the level. **Re-evaluation** at sprint cadence: classification is reviewed against current operating conditions; changes are propagated through the authority chain.

## 5. SKIPJACK COUPLING

Skipjack mechanisms operate over classified state.[11] **Context integrity** is enforced through classification verification at substrate boundaries. **Structured routing** uses classification as a routing-decision input (only substrates at appropriate classification levels are admissible). **HITL control points** fire on spillage detection. **Iterative feedback** closes via classification review at sprint ceremony.

## 6. IMPLICATIONS AND CONCLUSION

The implications extend to architecture (classification is a first-class distributed-state property), tooling (CDS components for cross-domain movement), operational discipline (classification reviews are scheduled), and compliance posture (classification integrity is auditable).

The core claim: **distributed-state classification integrity is the architectural commitment that makes governance-significant agentic operation feasible on the tactical substrate; without it, spillage is the dominant operational failure mode**.

Returned to $HGC^3AE^2$ letter by letter:

**H — Human-governed.** Classification policy and authority are human-led.

**G — Curated Context.** The classification registry is curated context.

**$C^1$ — Cybersecurity. Classification integrity is $C^1$** — load-bearing letter for this paper.

**$C^2$ — Continuity.** Continuity through classification preserved at substrate boundaries.

**$C^3$ — Coherence.** Classification metadata travels with state coherently across distribution.

**A — Agentically Engineered.** Classification enforcement is agentic within governance-defined policy.

**E — Executed.** Continuous classification verification at every state-movement event.

The exponent — the closure — is the assign-propagate-review loop.

10 Andrei Sabelfeld and Andrew C. Myers, “Language-Based Information-Flow Security,” *IEEE Journal on Selected Areas in Communications* 21, no. 1 (2003): 5–19, https://doi.org/10.1109/JSAC.2002.806121.

11 Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.

The remaining EDO papers (accreditation, interoperability, deployment) operate on the classification substrate this paper specifies.[12][13]

## REFERENCES

*— end v1.0-preprint-draft —*

**Drafting state:** v1.0-preprint draft. PR count: 12 inline (Bell-LaPadula MITRE + McLean SP · Biba MITRE · Hutchinson C&S · Solove CLSR · Liu TDSC · Sabelfeld JSAC · Newsome NDSS · Aleti ACM CSUR · Gelenbe FGCS · Rubin IEEE Computer · NIST + supporting). Existing draft drafts/edo-wave-2/P8-classification-problem.md folds into v0.5 expansion.

12 EDO-P9 *Accreditation Problem* promotion (in concurrent drafting; yks-build#108).

13 EDO-P11 CAPSTONE promotion (in concurrent drafting; yks-build#110).